Trigonometric Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 9$ in graphing tangent functions, evaluate $f(5)$.

Final Answer: 29. Substitute $x = 5$: $f(5) = 4(5) + 9 = 29$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 3$ in key features of secant, cosecant, and cotangent graphs, evaluate $f(2)$.

Final Answer: 13. Substitute $x = 2$: $f(2) = 5(2) + 3 = 13$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for key features of secant, cosecant, and cotangent graphs consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 16.5. The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(2)$.

Final Answer: 16. Substitute $x = 2$: $f(2) = 6(2) + 4 = 16$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for graphing secant, cosecant, and cotangent functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 17.5. The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The cosine function is symmetric about the y-axis because it is:.

Final Answer: Even. Applying Trigonometric Functions principles, $\cos(-x) = \cos x$: even function. Therefore, the verified correct option is Even.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Unit 3 · Trigonometric and Polar Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Prove the trigonometric identity $\frac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$ for all values of θ where both sides are defined.

Step 1: Substitute the double-angle identities $\cos 2\theta = 1 - 2\sin^2\theta$ and $\sin 2\theta = 2\sin\theta\cos\theta$: $\frac{1-(1-2\sin^2\theta)}{2\sin\theta\cos\theta}$ Step 2: Simplify the numerator: $1 - 1 + 2\sin^2\theta = 2\sin^2\theta$ Step 3: Form the ratio and cancel common factor $2\sin\theta$ (for $\sin\theta eq 0$): $\frac{2\sin^2\theta}{2\sin\theta\cos\theta} = \frac{\sin\theta}{\cos\theta} = \tan\boldsymbol{\theta}$ ■

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

Step 1: Factor the quadratic in $\sin x$: $(2\sin x - 1)(\sin x - 1) = 0$. Step 2: Solve each factor branch:Branch 1: $2\sin x - 1 = 0 \Rightarrow \sin x = \frac{1}{2}$. On $[0, 2\pi)$, $x = \frac{\pi}{6}$ (Quadrant I) or $x = \frac{5\pi}{6}$ (Quadrant II).Branch 2: $\sin x - 1 = 0 \Rightarrow \sin x = 1$. On $[0, 2\pi)$, $x = \frac{\pi}{2}$.The complete solution set is $x \in \left\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}\right\}$.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.

Expand the double-angle expressions on the left-hand side: LHS $= \frac{2\sin x\cos x}{\sin x} - \frac{2\cos^2 x - 1}{\cos x}$. Cancel $\sin x$ in the first term and combine over a common denominator of $\cos x$: $= 2\cos x - \frac{2\cos^2 x - 1}{\cos x} = \frac{2\cos^2 x - (2\cos^2 x - 1)}{\cos x}$. $= \frac{2\cos^2 x - 2\cos^2 x + 1}{\cos x} = \frac{1}{\cos x} = \sec x = \text{RHS}$. ✓

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Solve the trigonometric equation $2\cos^2 x - \sin x - 1 = 0$ for all x in the interval $[0, 2\pi)$.

Step 1: Apply the Pythagorean identity $\cos^2 x = 1 - \sin^2 x$: $2(1 - \sin^2 x) - \sin x - 1 = 0 \Rightarrow 2\sin^2 x + \sin x - 1 = 0$. Step 2: Factor the quadratic in $\sin x$: $(2\sin x - 1)(\sin x + 1) = 0 \Rightarrow \sin x = \frac{1}{2}$ or $\sin x = -1$. Step 3: Extract all solutions in $[0, 2\pi)$: $\boldsymbol{x \in \left\{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}\right\}}$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Prove the trigonometric identity $\frac{\sin(2x)}{1+\cos(2x)} = \tan x$ and state the domain exclusions.

Step 1: Apply double-angle identities $\sin(2x) = 2\sin x\cos x$ and $\cos(2x) = 2\cos^2 x - 1$: LHS $= \frac{2\sin x\cos x}{1 + (2\cos^2 x - 1)} = \frac{2\sin x\cos x}{2\cos^2 x}$. Step 2: Cancel $2\cos x$ (for $\cos x eq 0$): $\frac{\sin x}{\cos x} = \tan x = \text{RHS}$. ■ Domain exclusions: $\boldsymbol{x eq \frac{\pi}{2} + k\pi}$ for $k \in \mathbb{Z}$.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{3x+4}{2x-5}$, and state the domain of f^{-1}.

Let $y = \frac{3x+4}{2x-5}$. Swap variables: $x = \frac{3y+4}{2y-5}$. Multiply both sides by $2y-5$: $x(2y-5) = 3y+4 \Rightarrow 2xy - 5x = 3y + 4$. Group all y-terms on one side: $2xy - 3y = 5x + 4 \Rightarrow y(2x-3) = 5x+4$. Divide to isolate y: $f^{-1}(x) = \frac{5x+4}{2x-3}$. Domain of f^{-1}: $x \neq \frac{3}{2}$ (all real numbers except $x = 1.5$).

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given $f(x) = \frac{2}{x-1}$ and $g(x) = \sqrt{x+5}$, find (a) the value of $(f \circ g)(4)$, and (b) the domain of $g(x)$.

1. Evaluate the inner function first: $g(4) = \sqrt{4+5} = \sqrt{9} = 3$. 2. Substitute into f: $(f \circ g)(4) = f(g(4)) = f(3) = \frac{2}{3-1} = \frac{2}{2} = 1$. 3. Domain of $g(x) = \sqrt{x+5}$: The radicand must be non-negative: $x + 5 \geq 0 \Rightarrow x \geq -5$ or in interval notation $[-5, \infty)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Polynomial Functions

Determine whether the function $g(x) = \frac{x^3}{x^2+4}$ is even, odd, or neither.

Evaluate $g(-x)$ by substituting $-x$ for every occurrence of x: $g(-x) = \frac{(-x)^3}{(-x)^2+4} = \frac{-x^3}{x^2+4} = -\left(\frac{x^3}{x^2+4}\right) = -g(x)$ Because $g(-x) = -g(x)$ for all $x \in \mathbb{R}$, $g(x)$ is an odd function (symmetric with respect to 180° rotation about the origin).

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Rational Expressions

If $f(x) = 2x - 5$ and $g(x) = x^2 + 3$, find the expression for $(f \circ g)(x)$ and solve for all x such that $(f \circ g)(x) = 25$.

Step 1: Form the composite function $(f \circ g)(x) = f(g(x))$: $f(g(x)) = f(x^2+3) = 2(x^2+3) - 5 = 2x^2 + 6 - 5 = \mathbf{2x^2 + 1}$ Step 2: Solve $2x^2 + 1 = 25$: $2x^2 = 24 \Rightarrow x^2 = 12 \Rightarrow x = \pm\sqrt{12} = \mathbf{\pm 2\sqrt{3}}$

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A function is defined piecewise: $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$. What is the value of $f(-3) + f(2)$?

Evaluate each term according to its interval condition: For $x = -3 < 0$: $f(-3) = (-3)^2 = 9$. For $x = 2 \geq 0$: $f(2) = 2(2) + 1 = 5$. Sum: $f(-3) + f(2) = 9 + 5 = \mathbf{14}$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Find all solutions to $2x^2 - 6x + 5 = 0$ using the quadratic formula, and determine whether the roots are real or complex.

Identify coefficients: $a = 2$, $b = -6$, $c = 5$. Evaluate the discriminant:
$\Delta = b^2 - 4ac = (-6)^2 - 4(2)(5) = 36 - 40 = -4$. Because \(\Delta

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Sequences & Series

[Functions & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Conic Sections

[Quadratic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(2)$.

a. 11 ✓ **b.** 13

c. 4 **d.** 15

Substitute $x = 2$: $f(2) = 2(2) + 7 = 11$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 8$ in graphing sine and cosine functions, evaluate $f(2)$.

a. 14 ✓ **b.** 17

c. 6 **d.** 18

Substitute $x = 2$: $f(2) = 3(2) + 8 = 14$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for key features of sine, cosine, and tangent graphs consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5 ✓

d. 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for graphing sine and cosine functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for graphing tangent functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the maximum value of the function $f(x) = \sqrt{3}\sin(2x) - \cos(2x) - 2\sin x + 2\sqrt{3}\cos x$ for all real numbers x?

a. **3** ✓

b. 2

c. 6

d. 1

We solve the problem by synthesizing harmonic form (linear combination of sine and cosine), angle-doubling relationships, and quadratic optimization on a restricted domain. **Step 1: Express the single-angle terms in harmonic form.** Recall the linear combination identity $A\cos x + B\sin x = R\cos(x - \alpha)$. For the terms $2\sqrt{3}\cos x - 2\sin x$: $2\sqrt{3}\cos x - 2\sin x = 4\left(\frac{\sqrt{3}}{2}\cos x - \frac{1}{2}\sin x\right)$ Using the cosine addition formula.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = 4\cos^2(x) + b\sin(x) + c$ where b and c are real constants with $b > 0$. If the maximum value of $f(x)$ is 10 and the minimum value of $f(x)$ is 1, what is the value of c?

a. 5 ✓

b. $\frac{11}{2}$

c. 9

d. 4

To find the unknown parameters, we first express $f(x)$ in terms of a single trigonometric function using the identity $\cos^2(x) = 1 - \sin^2(x)$: $f(x) = 4(1 - \sin^2(x)) + b\sin(x) + c = -4\sin^2(x) + b\sin(x) + c + 4$ Let $u = \sin(x)$. Since $x \in \mathbb{R}$, the variable u is restricted to the interval $[-1, 1]$. We analyze the quadratic function: $g(u) = -4u^2 + bu + c + 4, \quad u \in [-1, 1]$ The graph of $g(u)$ is a downward-opening parabola with its vertex at: $u_v = -\frac{b}{2(-4)} = \frac{b}{8}$ S.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The function $f(x) = a\cos(bx) + d$, where $a, b,$ and d are positive constants, has a period of π, a maximum value of 7, and satisfies $f(\frac{\pi}{6}) = 4$. What is the value of $f(\frac{\pi}{3})$?

a. -2 ✓

b. 1

c. 4

d. -5

We determine the unknown parameters $a, b,$ and d step by step: 1. **Find b using the period:** The period of $f(x) = a\cos(bx) + d$ is given by $\frac{2\pi}{b}$. Setting this equal to π: $\frac{2\pi}{b} = \pi \Rightarrow b = 2$ 2. **Use the maximum value to relate a and d:** Since $a > 0$, the maximum value occurs when $\cos(bx) = 1$: $a(1) + d = 7 \Rightarrow a + d = 7$ 3. **Use the given point $f(\frac{\pi}{6}) = 4$:** Substitute $b = 2$ and $x = \frac{\pi}{6}$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

In a right-angled triangle, one of the acute angles is 30°. What is the measure of the other acute angle?

a. 30°

b. 45°

c. 60° ✓

d. 90°

In a right-angled triangle, the sum of the three angles is 180°. One angle is 90° (the right angle), and another acute angle is given as 30°. Let the other acute angle be x. Therefore, we have the equation: $90^\circ + 30^\circ + x = 180^\circ$. Subtracting 120° from both sides gives $x = 180^\circ - 120^\circ = 60^\circ$.

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

State and prove De Moivre's theorem: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

a. **By induction: base trivial; step uses** $(\cos\theta + i\sin\theta)^{n+1} = (\cos n\theta + i\sin n\theta)(\cos\theta + i\sin\theta) = \cos(n+1)\theta + i\sin(n+1)\theta$**.** ✓ ✓

b. Not provable by induction

c. Only for $n \in \mathbb{Z}^+$

d. Use Euler's formula directly

Applying Trigonometric Functions principles, Standard proof by induction. Therefore, the verified correct option is By induction: base trivial; step uses $(\cos\theta + i\sin\theta)^{n+1} = (\cos n\theta + i\sin n\theta)(\cos\theta + i\sin\theta) = \cos(n+1)\theta + i\sin(n+1)\theta$. ✓.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Prove the product formula: $\cos A\cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)]$.

a. Only if $A = B$

b. Use integration

c. Not true

d. **From $\cos(A-B) = \cos A\cos B + \sin A\sin B$ and $\cos(A+B) = \cos A\cos B - \sin A\sin B$. Add and halve. ✓ ✓**

Applying Trigonometric Functions principles, Standard product-to-sum. Therefore, the verified correct option is From $\cos(A-B) = \cos A\cos B + \sin A\sin B$ and $\cos(A+B) = \cos A\cos B - \sin A\sin B$. Add and halve. ✓.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Prove the sum-to-product formula: $\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$.

a. Not true

b. Only for $A = B$

c. **Set $A = M+N, B = M-N$; then $\sin(M+N)+\sin(M-N) = 2\sin M\cos N$. Substitute $M = (A+B)/2, N = (A-B)/2$. ✓ ✓**

d. Use calculus

Applying Trigonometric Functions principles, Standard derivation. Therefore, the verified correct option is Set $A = M+N, B = M-N$; then $\sin(M+N)+\sin(M-N) = 2\sin M\cos N$. Substitute $M = (A+B)/2, N = (A-B)/2$. ✓.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Prove the identity $\sin 3A = 3\sin A - 4\sin^3 A$ using the angle addition formula.

a. Not an identity

b. Only for $A < \pi/2$

c. Use calculus

d. $\sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A = 3\sin A - 4\sin^3 A$. ✓ ✓

Applying Trigonometric Functions principles, Standard derivation. Therefore, the verified correct option is $\sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A = 3\sin A - 4\sin^3 A$. ✓.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find all values of θ such that $\sin\theta, \cos\theta, \tan\theta$ are in arithmetic progression.

a. $\theta \approx 0.6435$ **rad (a special angle satisfying** $2\cos\theta = \sin\theta + \tan\theta$**)** ✓

b. $\theta = \pi/3$

c. No solution exists

d. $\theta = \pi/4$

$2\cos\theta = \sin\theta + \tan\theta \Rightarrow 2\cos^2\theta\sin\theta = \sin^2\theta + \sin^2\theta/\cos\theta \cdot \cos\theta$. Complex transcendental equation.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the half-angle formula for $\cos(\theta/2)$.

a. $\pm\sqrt{\frac{1+\cos\theta}{2}}$ ✓

b. $\cos\theta/2$

c. $\pm\sqrt{\frac{1-\cos\theta}{2}}$

d. $\sqrt{1+\cos\theta}$

Applying Trigonometric Functions principles, Half-angle formula for cosine. Therefore, the verified correct option is $\pm\sqrt{\frac{1+\cos\theta}{2}}$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

State the sum-to-product formula for $\cos A - \cos B$.

a. $2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$

b. $2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$

c. $2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$

d. $-2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$ ✓

Applying Trigonometric Functions principles, Standard sum-to-product. Therefore, the verified correct option is $-2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Using the product-to-sum formula, simplify $\sin A\cos B$.

a. $\cos(A-B)$

b. $\frac{1}{2}[\sin(A+B)+\sin(A-B)]$ ✓

c. $\sin(A+B)$

d. $\frac{1}{2}[\cos(A-B)-\cos(A+B)]$

Applying Trigonometric Functions principles, Product-to-sum formula. Therefore, the verified correct option is $\frac{1}{2}[\sin(A+B)+\sin(A-B)]$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Evaluate $\sin 3\theta$ in terms of $\sin\theta$.

a. $3\sin\theta - 4\sin^3\theta$ ✓

b. $4\sin^3\theta - 3\sin\theta$

c. $3\sin\theta + 4\sin^3\theta$

d. $3\sin\theta\cos^2\theta$

Applying Trigonometric Functions principles, Triple angle formula for sine. Therefore, the verified correct option is $3\sin\theta - 4\sin^3\theta$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Sequences & Series

Find the inverse of $y = \cos x$ on $[0, \pi]$.

a. $x = \arcsin y$

b. $x = \pi - y$

c. $x = 2\pi - \arccos y$

d. $x = \arccos y$ **for** $y \in [-1, 1]$ ✓

Applying Trigonometric Functions principles, Principal branch of arccos. Therefore, the verified correct option is $x = \arccos y$ for $y \in [-1, 1]$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Conic Sections

Find the inverse function: if $y = \sin x$, then $x =$

a. $\sin^{-1} y$ for $y \in [0, 1]$ only

b. $\cos^{-1} y$

c. π/y

d. $\arcsin y$ **for** $y \in [-1, 1]$ ✓

Applying Trigonometric Functions principles, Definition of arcsine. Therefore, the verified correct option is $\arcsin y$ for $y \in [-1, 1]$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Functions & Their Graphs

State the angle addition formula $\sin(A + B) =$

a. $\cos A \cos B - \sin A \sin B$

b. $\sin A \cos B + \cos A \sin B$ ✓

c. $\sin A \cos B - \cos A \sin B$

d. $\sin A + \sin B$

Applying Trigonometric Functions principles, Angle addition formula. Therefore, the verified correct option is $\sin A \cos B + \cos A \sin B$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Matrices & Systems

State the double-angle formula $\sin 2\theta =$

a. $2\cos^2 \theta - 1$

b. $\sin^2 \theta - \cos^2 \theta$

c. $2\sin \theta \cos \theta$ ✓

d. $\cos^2 \theta - \sin^2 \theta$

Applying Trigonometric Functions principles, Double angle formula. Therefore, the verified correct option is $2\sin \theta \cos \theta$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Probability & Statistics

Find the reference angle for 210°.

a. 45°

b. 60°

c. 150°

d. 30° ✓

Applying Trigonometric Functions principles, $210^\circ - 180^\circ = 30^\circ$. Therefore, the verified correct option is 30°.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Convert $\pi/3$ radians to degrees.

a. 45°

b. 60° ✓

c. 90°

d. 30°

Applying Trigonometric Functions principles, $\pi/3 \times 180/\pi = 60^\circ$. Therefore, the verified correct option is 60°.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Polynomial Functions

For $y = \sin x$, the x-intercepts occur at:

a. $x = (2n + 1)\pi/2$

b. $x = 2n\pi$

c. $x = n\pi/2$

d. $x = n\pi,\ n \in \mathbb{Z}$ ✓

Applying Trigonometric Functions principles, $\sin(n\pi) = 0$. Therefore, the verified correct option is $x = n\pi,\ n \in \mathbb{Z}$.

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